

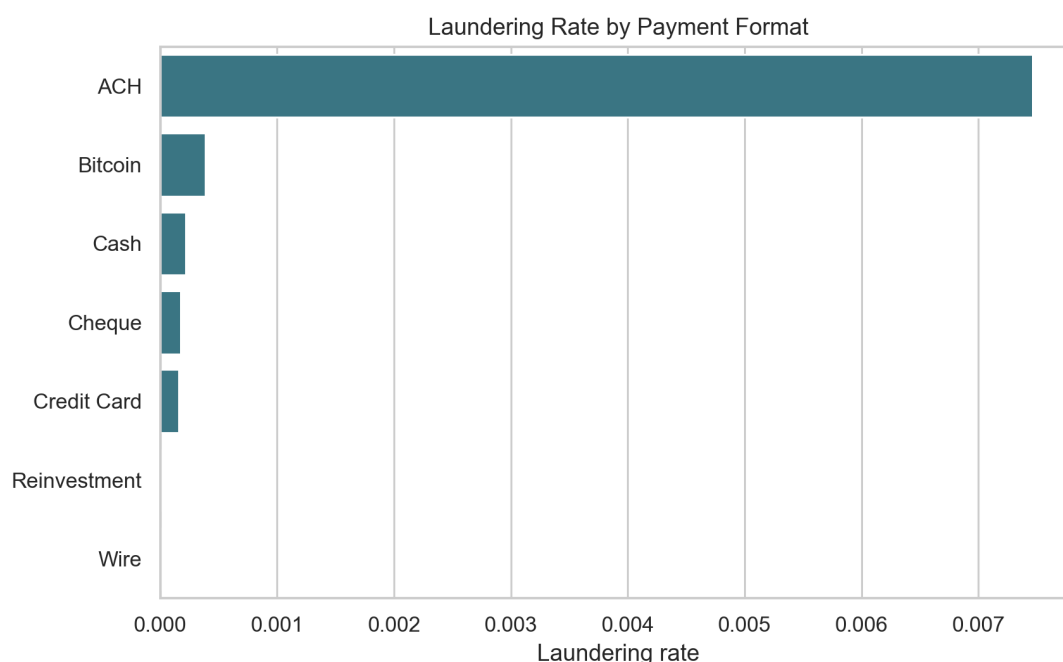
Detecting Money Laundering in Synthetic Financial Transaction Networks

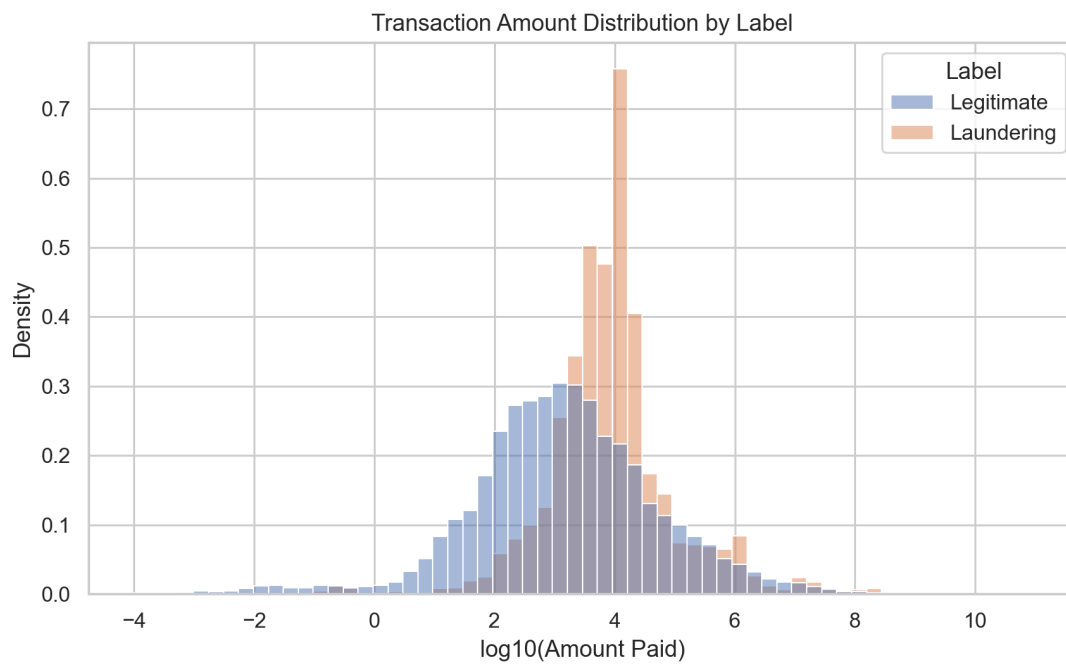
From tabular ML to graph-aware AML features

The full dataset contains 5,078,345 transactions, including 5,177 laundering transactions (0.1019%). This rare positive class makes accuracy insufficient, so the analysis emphasizes recall, F1, ROC-AUC, and PR-AUC.

Dataset and preprocessing

We engineered temporal, amount, transaction, and graph-aware account behavior features. Accounts were represented as nodes and transactions as directed edges. Sender out-degree, receiver in-degree, unique counterparties, total sent/received amount, and PageRank were added as open-ended graph features.

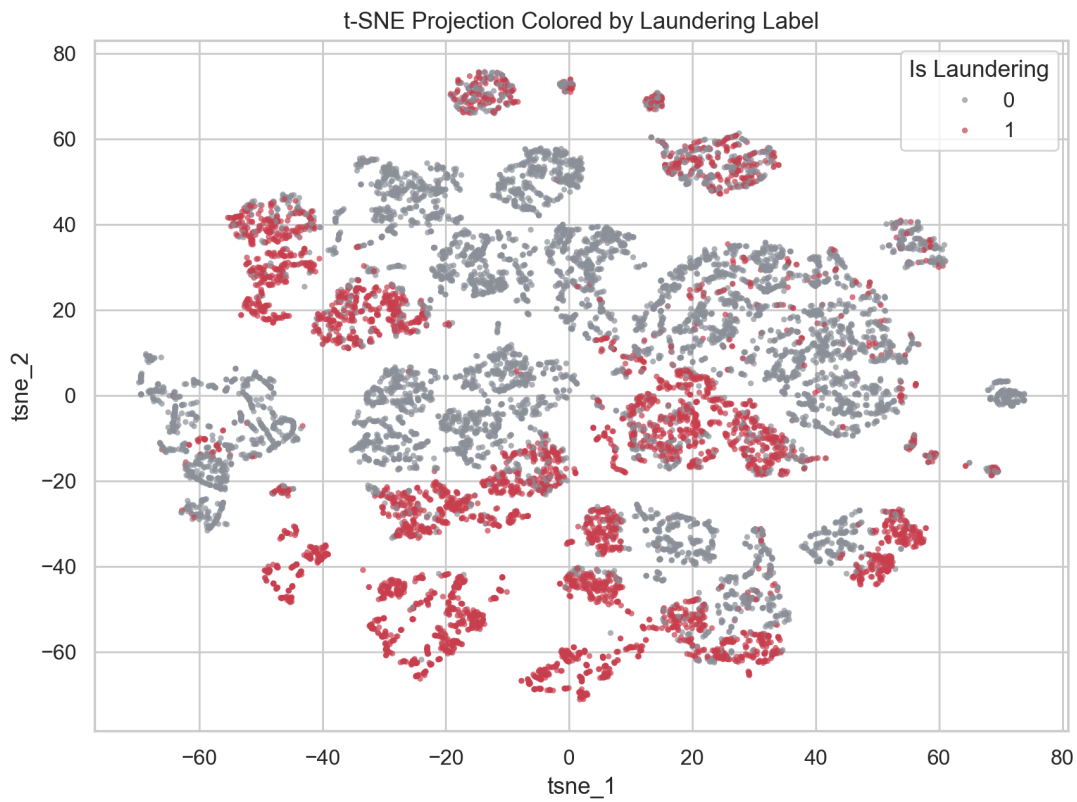


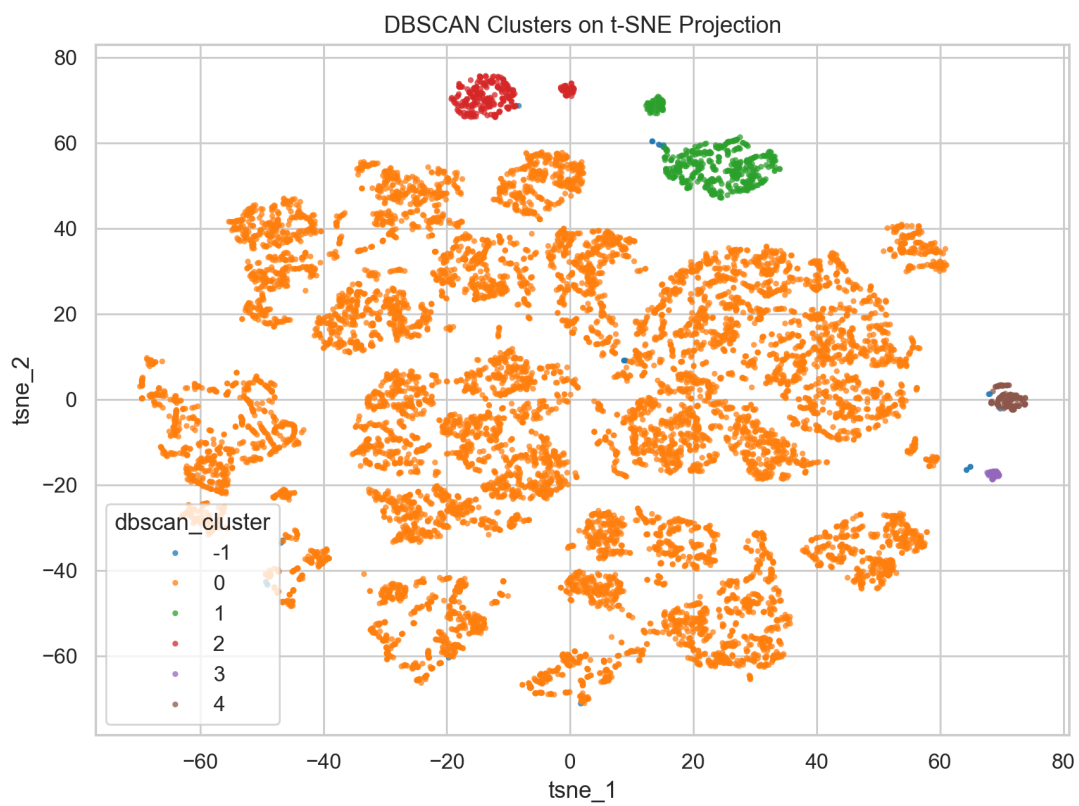


Visualization and clustering

A stratified sample was projected with t-SNE and clustered with MiniBatch K-Means and DBSCAN. DBSCAN produced the stronger Silhouette score, while K-Means exposed a high-risk cluster in the enriched visualization sample.

Method	Clusters	Noise	Silhouette	DB	CH
MiniBatchKMeans	6	0.000	0.145	1.755	1274.1
DBSCAN	5	0.002	0.324	0.767	1437.8

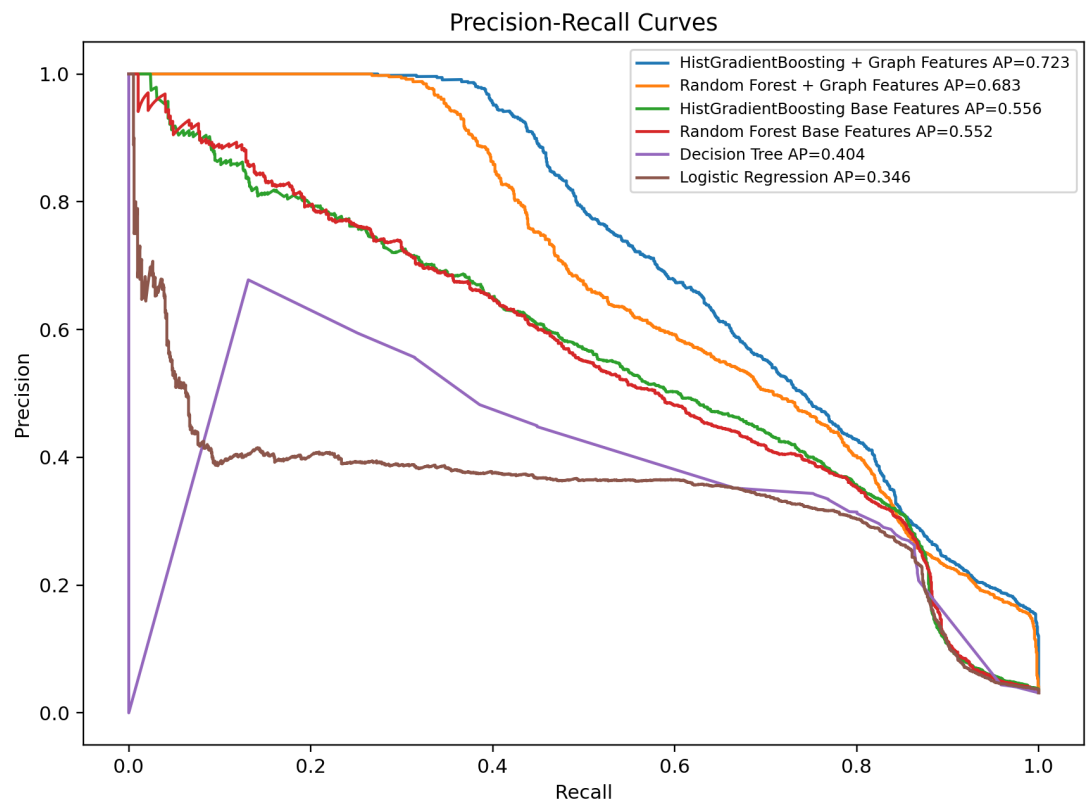


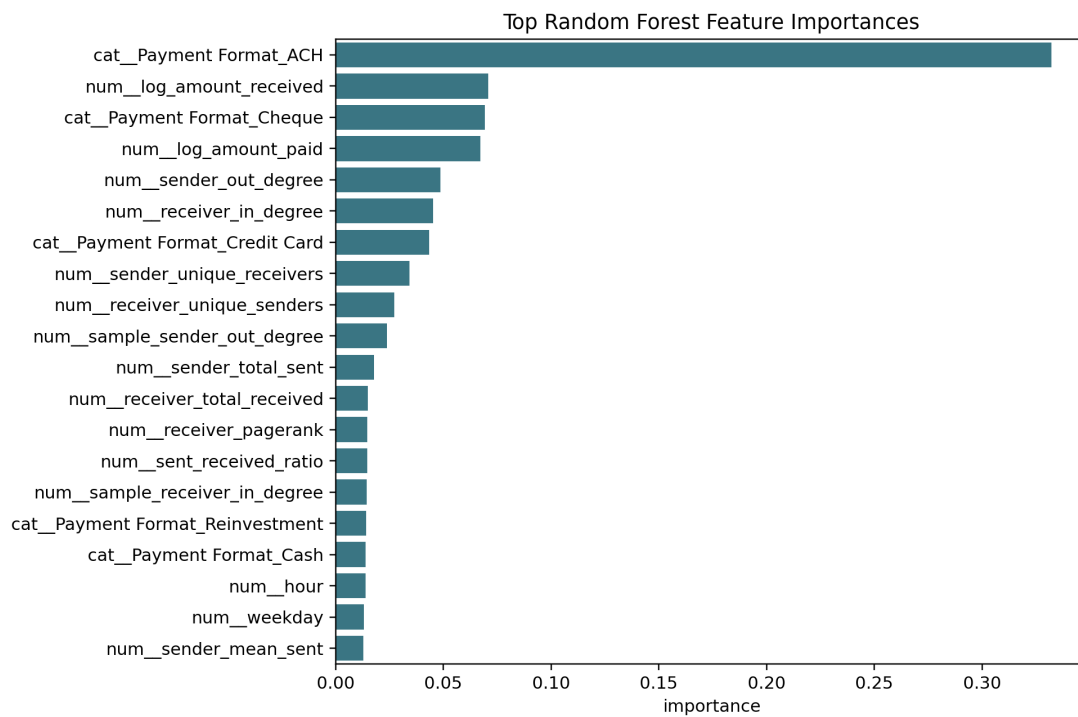


Prediction and evaluation

Model	Prec.	Recall	F1	ROC	PR
HistGradientBoosting + Graph Features	0.658	0.627	0.642	0.976	0.723
Random Forest + Graph Features	0.562	0.639	0.598	0.972	0.683
HistGradientBoosting Base Features	0.490	0.623	0.548	0.931	0.556
Random Forest Base Features	0.505	0.578	0.539	0.929	0.552
Decision Tree	0.341	0.755	0.470	0.911	0.404
Logistic Regression	0.352	0.670	0.461	0.915	0.346

The best model is HistGradientBoosting with graph features. Compared with base HistGradientBoosting, graph features improve F1 from 0.548 to 0.642 and PR-AUC from 0.556 to 0.723.





Conclusion

Simple tabular models are useful baselines, but graph-aware behavioral features substantially improve minority-class detection. This suggests that laundering is better modeled as network behavior rather than isolated row-level anomalies. Limitations include synthetic data, sampling, and the need for time-based validation in future work.

References

IBM AML-Data GitHub repository.

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